

**PROJECT TITLE:** DATA SCIENCE PROJECT (Trading Behavior vs Market Sentiment Analysis)

**CANDIDATE NAME:** J. Supritha

**DATE:** 27-11-2025

## **1. Abstract:**

This project examines how market psychology, represented through the Bitcoin Fear & Greed index, influences trading behavior on the Hyperliquid platform. By merging sentiment classification with granular trade-level records, the study identifies how leverage usage, position sizing, profitability, and risk exposure shift during Fear versus Greed phases. The analysis reveals that enthusiasm during Greed phases encourages riskier strategies, including increased leverage and larger sizes, yet these behaviors do not consistently translate into improved performance. Fear phases, while more conservative, display steadier PnL outcomes driven by disciplined exposure. This results indicate that integrating sentiment awareness into trading systems may significantly enhance decision-making and reduce unnecessary volatility.

Beyond uncovering behavioral tendencies, this project highlights the value of sentiment-driven analytics in risk-controlled trading. The results suggest that a purely technical or price-action-based approach overlooks an important psychological variable that shapes trader outcomes. By blending sentiment trades with quantitative trade metrics, trading systems can forecast periods of elevated risk-taking and react accordingly. This creates an opportunity to build smarter, adaptive trading frameworks that protect capital during emotionally charged market conditions while exploiting opportunities when sentiment aligns with disciplined strategy.

## **2. Introduction:**

Financial markets are profoundly shaped by collective sentiment, especially in volatile sectors like cryptocurrency. Traders often make decisions driven not only by rational analysis but also by emotional reactions to price movement and market hype. Understanding how emotional sentiment correlates with tangible trading outcomes can strengthen risk management and reveal exploitable behavioral patterns.

This work explores whether traders on Hyperliquid follow or contradict broader market sentiment trends and how such alignment affects profitability. By comparing risk exposure and trading results across Fear and Greed market states, the study aims to determine if sentiment acts as a predictive input for smarter trade execution.

Evaluating sentiment from an empirical standpoint is especially important in cryptocurrency because its participants are predominantly retail traders who tend to be more emotionally reactive than institutional actors. Unlike traditional markets where regulations and long-standing valuation benchmarks influence decisions, crypto trading behavior is driven largely by perception and hype cycles. Measuring whether traders contribute to or are affected by emotional market swings provides a practical basis for designing safeguards that align with real-world behavior rather than theoretical discipline. This understanding creates the foundation for sentiment-aware decision systems that support both discretionary and algorithmic traders.

### **3. DataSet Description:**

Two datasets form the basis of this research. The Bitcoin Market Sentiment dataset provides daily sentiment labels- Fear or Greed-reflecting prevailing investor psychology. The second dataset contains detailed historical trades from the Hyperliquid exchange, including fields such as account ID, execution price, trade size, direction, leverage, position status, and close PnL.

To analyze trader reactions to sentiment shifts, both datasets were merged on matching execution dates. Additional calculated features- including profitability labels, daily volume, and risk scores – enhanced interpretability and supported deeper behavioral pattern discovery.

The dataset's combination provides a unique multi-dimensional view of both market psychology and individual trader decisions. The sentiment dataset reflects the emotional environment of the overall market, while the Hyperliquid dataset demonstrates how individual actors respond under those emotional conditions. This pairing enables not only macro-to-micro comparison but also deeper segmentation across trader types, time windows, and levels of exposure. The presence of high-frequency timestamps also allows temporal analysis to determine whether rapid sentiment changes lead to momentum-driven trading spikes or panic-driven exit patterns.

## **4. Data preprocessing:**

Several preparation steps were undertaken to make the datasets suitable for analysis. Time values in the Hyperliquid dataset were converted to dates to ensure alignment with the sentiment index. Duplicate and incomplete trades were removed, and missing values were treated according to column relevance to avoid data loss or misleading noise. Outliers in PnL and leverage were detected and capped to stabilize statistical outcomes.

Feature engineering played a crucial role: new variables were introduced to represent profitability categories, risk levels, and aggregate trader metrics per day. Once both datasets were cleaned and enhanced, they were merged to associate every trade with its corresponding sentiment state.

Special attention was given to preserving the integrity of the dataset while minimizing noise that could distort interpretation. Instead of discarding high-risk trades outright, corrective treatments ensured that legitimate but extreme executions remained part of the analysis because they reflect real trader psychology during market stress or enthusiasm. Additionally, feature engineering allowed the derivation of normalized metrics that enable fair comparison between large accounts and small accounts, ensuring that insights reflect behavioral patterns rather than merely differences in capital size. These preprocessing steps built a stable foundation for meaningful statistical and visual exploration.

## **5. Exploratory Data Analysis(EDA):**

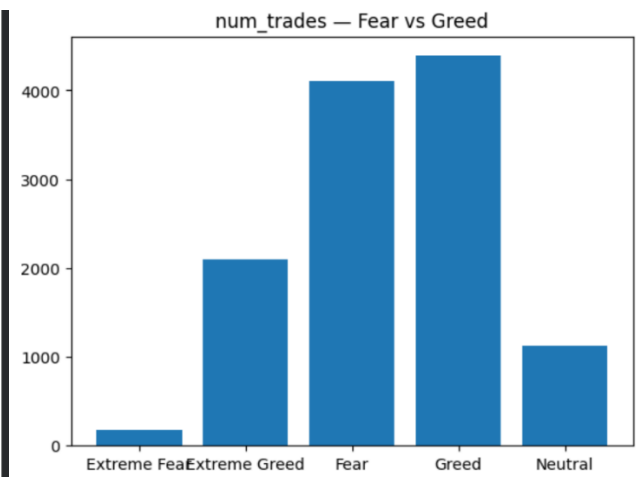
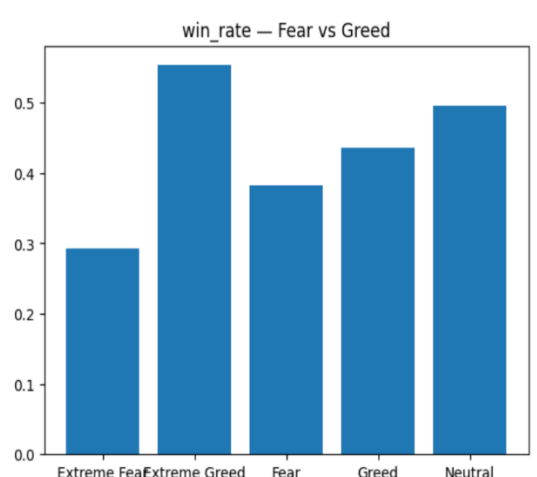
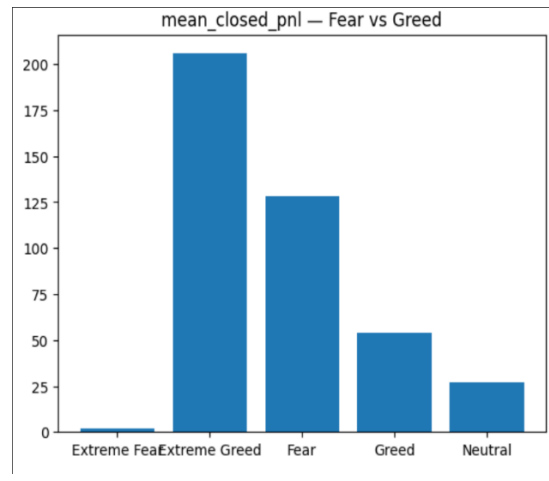
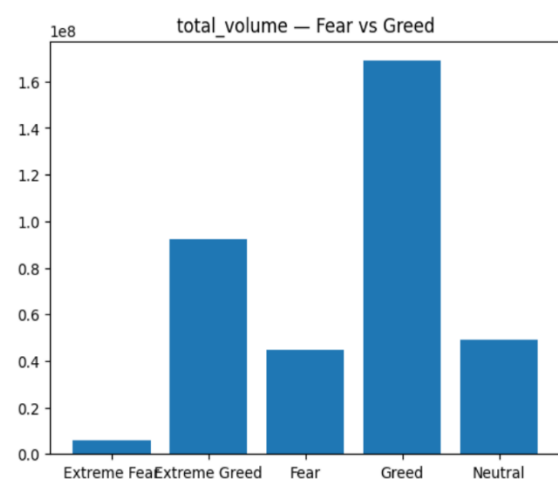
EDA revealed distinct shifts in trader psychology depending on market sentiment. During Greed phases, traders frequently adopted larger positions, applied higher leverage, and demonstrated increased trade volume-showing confidence and risk appetite. In contrast, Fear periods showed reduced trading activity and smaller exposure levels, signaling defensive positioning.

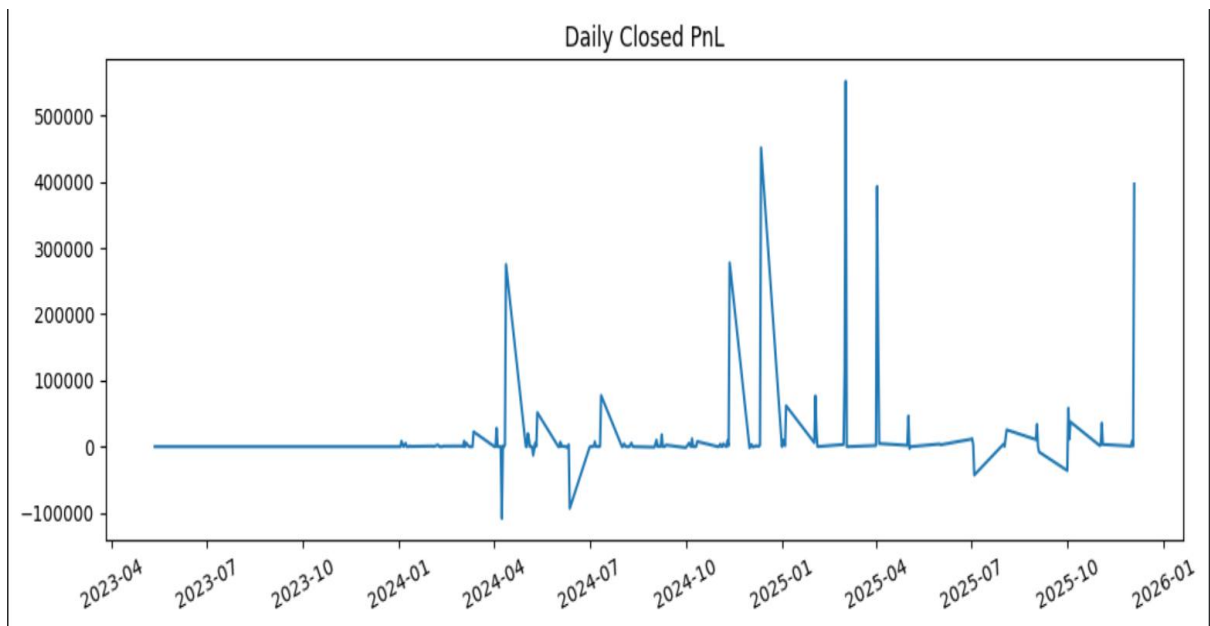
Despite the higher activity during Greed, PnL volatility increased substantially, suggesting that elevated confidence does not guarantee superior returns. Conversely, Fear periods produced more stable PnL performance due to controlled risk. The visualizations stored in the /outputs folder highlight these behavioral contrasts clearly.

Patterns observed in EDA also point toward the possibility that certain trader segments behave differently depending on sentiment. Some accounts appear to exploit Greed sentiment to aggressively scale profitable positions, while others repeatedly overextend during hype periods and incur substantial losses. Conversely, cautious traders seem more consistent but may miss opportunities when markets transition out of Fear back into neutral or bullish conditions. These divergences reveal that not all traders respond uniformly to sentiment stimuli and that recognizing trader-specific sensitivity to sentiment could be essential in designing individualized strategy recommendations.

## 6. Output:

|                | num_trades | total_volume | mean_closed_pnl | median_closed_pnl | win_rate | avg_leverage |
|----------------|------------|--------------|-----------------|-------------------|----------|--------------|
| classification |            |              |                 |                   |          |              |
| Extreme Fear   | 174.0      | 6.112058e+04 | 21.455587       | 0.0               | 0.252874 | NaN          |
| Extreme Greed  | 2095.0     | 7.392020e+07 | 456.166481      | 0.0               | 0.457757 | NaN          |
| Fear           | 4110.0     | 3.731878e+07 | 341.731718      | 0.0               | 0.475912 | NaN          |
| Greed          | 4388.0     | 1.581985e+08 | 55.759296       | 0.0               | 0.412261 | NaN          |
| Neutral        | 1120.0     | 4.601550e+07 | 37.320966       | 0.0               | 0.450000 | NaN          |







## 7. Key Insights:

- Risk-seeking behavior rises sharply during Greed sentiment without proportional improvement in profitability.
- Fear Sentiment encourages disciplined trading with reduced risk but yields steadier results over time.
- Traders appear to follow market psychology rather than act contrarian, increasing risk during bullish confidence and scaling back in fearful phases.
- Sentiment-aware strategies have potential to prevent over-leveraging in Greed phases and capitalize on stable returns during Fear Phases.

Another important insight is that trader outcomes depend less on sentiment itself and more on how traders respond to it. Greed sentiment does not inherently lead to losses; instead, undisciplined scaling and leverage amplification during such periods increase volatility. Similarly, Fear does not prevent profitability; rather, scarcity of participation causes traders to overlook structured opportunities. This distinction reinforces the idea that emotional conditions amplify decision biases, and managing these biases is more impactful than simply forecasting sentiment direction.

## **8. Conclusion:**

Sentiment is a meaningful indicator of how traders behave and how profits fluctuate. The study shows that Greed tends to push traders toward aggressive strategies that can amplify both gains and losses, while Fear enforces caution, often leading to more predictable PnL Performance. Incorporating sentiment signals into real-time trading decisions can support improved risk management and foster more consistent profitability.

Ultimately, the results highlight a crucial principle: the success of a trade depends not only on market direction but also on the trader's psychological resilience in the face of external sentiment. The study demonstrates that the most significant risks may arise not from market volatility but from human overconfidence and panic. By acknowledging the role of emotion in trade execution and integrating sentiment into strategic evaluation, traders and institutions can create more reliable frameworks capable of navigating both euphoric and fearful market environments.